Project Abstract

Intelligent Thermal Monitoring and Protection System for Electronic Circuits Using Embedded Systems

Abstract: In the era of Industry 4.0, predictive maintenance is vital for ensuring operational efficiency and safety. This project presents a robust embedded system designed to monitor the health of factory machinery in real-time. Unlike simulation-based models, this implementation utilizes a physical hardware layer powered by the ESP32 microcontroller, integrated with ACS712 current sensors and DS18B20 temperature probes to acquire live telemetry data directly from the equipment.

The ESP32 processes raw sensor signals and transmits the data via Wi-Fi (HTTP Protocol) to a centralized Node.js server. The server acts as the control hub, employing a WebSocket architecture to push live updates to a "Glassmorphism" styled web dashboard. This allows operators to visualize machine performance (Current and Temperature) with zero latency.

Key features include an automated safety logic system that continuously evaluates incoming data against safety thresholds. If a critical anomaly (e.g., Overheating > 90°C) is detected, the system triggers an immediate alert via Nodemailer, dispatching a detailed HTML diagnostic report to maintenance supervisors. Furthermore, the dashboard provides interactive analytics with historical trend visualization and downloadable PDF maintenance reports powered by jsPDF. This end-to-end solution—from hardware sensors to cloud-based visualization—offers a cost-effective, scalable approach to preventing machine failure and ensuring industrial safety.